Served stats

Throughput, latency and quality of each served recipe, by model and hardware

2026-10-16 06:00 UTC

Rendered 2026-10-16 06:00 UTC from stats/*.json, the ledger every harness run and harness export adds to (python3 -m harness stats renders it again). Aggregate tok/s, greedy, 256-token replies, the recipe's own ruler; engine rounds from the server's own counters.

Overview: by model and hardware

Aggregate tok/s as prose/code/structured/list; first token in seconds (one stream / the most streams).

model	hardware	one stream	most streams: total	first token (s)	suite	measured
GLM-5.3-Flash, EXL3 4 bpw	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	46/69/57/59		0.18	tools 59/60, json 30/30, gsm8k 246/250, humaneval 159/164, ifeval 118/150	2026-10-14
GLM-5.3-Flash, MLX 4-bit	1x Apple M5 Ultra (256 GB unified), macOS	91/104/99/42		0.12		2026-10-10
Gemma 4 31B, MLX 4-bit + DFlash	1x Apple M5 Ultra (256 GB unified), macOS	90/154/148/126		0.11		2026-10-15
NVIDIA Nemotron 3.5 Bolt 74B-A9B (24 GB: IQ2_XXS experts, Q8_0, NVFP4 Mamba) + vision	1x NVIDIA RTX 5090 (32 GB)	150/150/150/150		0.03	tools 58/60, json 30/30, gsm8k 241/250, humaneval 153/164, ifeval 120/150	2026-10-15
Qwen3.8-27B, MLX 4-bit + DFlash2	1x NVIDIA RTX 5090 (32 GB)	305 (median of 4 prompts)		0.10	tools 59/60, json 30/30, gsm8k 242/250, humaneval 153/164, ifeval 124/150	2026-10-14
Qwen3.8-Flash-Next INT4-Mixed (AutoRound)	2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE	77/127/101/130	16: 302/510/391/519	0.03 / 0.44-0.55	tools 60/60, json 30/30, gsm8k 241/250, humaneval 157/164, ifeval 130/150	2026-10-15
Qwen3.8-Flash-Next, 6-bit with vision	1x Apple M5 Ultra (256 GB unified), macOS	128/174/148/175	8: 272/330/292/339	0.16 / 0.76-0.84	tools 59/60, json 30/30	2026-10-12
Qwen3.8-Flash-Next, MLX 6-bit	1x Apple M5 Ultra (256 GB unified), macOS	145/206/172/194	4: 566/816/673/854	0.09 / 1.63-3.07	gsm8k 242/250, ifeval 134/150	2026-10-11
Step-5-Preview, MLX 2-bit experts (8-bit rest) + MTP	1x Apple M5 Ultra (256 GB unified), macOS	39/51/41/48		0.24	tools 58/60, json 30/30, gsm8k 49/50, ifeval 41/50	2026-10-15

Gemma 4 31B, MLX 4-bit + DFlash on 1x Apple M5 Ultra (256 GB unified), macOS

Target gemma4-31b-m5: Gemma 4 31B 4-bit (groups of 64), DFlash drafts, images, TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as gemma4-31b. Latest: run 20261015-200951-quick (2026-10-15 20:09, plan quick, done).

Serving: engine_commit 8c37d22c; binary_sha256 bb91242a259030711be1d274c53dbbfbcca01d805d94c2cc6de4da1c07919e2b; served_model {"object": "list", "data": [{"id": "gemma4-31b", "object": "model", "created": 1792113009,; chip Apple M5 Ultra; 274877906944

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	90.4 (90.4)	153.8 (153.8)	148.4 (148.4)	126.1 (126.1)	0.11-0.15 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.315	13.036	18%	25.57	12

Ruler-after

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	90.1 (90.1)	164.2 (164.2)	145.9 (145.9)	124.2 (124.2)	0.10-0.16 s

Engine rounds (ruler-after):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.328	12.662	18%	25.39	17

Suite

check	passed
soak	60/60
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	8/8
stop	3/3

check	passed
thinking	10/10
copy	2/2
max-tokens	3/3

Checks

check	result
memory-idle	measured: mac: System-wide memory free percentage: 96%; total = 4096.00M us
structured	FAIL: 0/18 obey; a bad regex -> 400; failed json_object/greedy, json_object/sampled, json_object/thinking, json_schema/greedy, json_schema/sampled, json_schema/thinking, guided_json/greedy, guided_json/sampled, guided_json/thinking, regex/greedy, regex/sampled, regex/thinking, choice/greedy, choice/sampled, choice/thinking, grammar/greedy, grammar/sampled, grammar/thinking
sampling	pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors
turns	pass: resumed == again == cold: True; cached 5494 of 5564; ttft resumed 0.135s, cold 0.134s

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-15 20:09	quick	done	153.8			114/114

GLM-5.3-Flash, EXL3 4 bpw on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 over RoCE

Target glm53-exl3-sparks: GLM-5.3-Flash EXL3 4 bpw, TensorFold native, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA; served as GLM-5.3-Flash-EXL3. Latest: run evidence/2026-10-14 (2026-10-14 00:00, plan published evidence, published).

Ruler-shipped-14fb445e-c1-sampled

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	45.1 (45.2)	64.6 (65.4)	56.5 (56.2)	57.0 (54.9)	0.18-0.25 s

Ruler-shipped-14fb445e-c1

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	46.5 (45.4)	68.7 (69.4)	57.0 (56.4)	58.8 (55.3)	0.18-0.24 s

Ruler-shipped-14fb445e-c16

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
16	144.0 (8.9)	198.3 (12.1)	180.1 (11.3)	208.4 (12.5)	0.91-1.19 s

Ruler-shipped-14fb445e-c8

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
8	110.1 (13.6)	152.5 (18.6)	138.7 (17.5)	157.0 (18.7)	0.49-0.63 s

Ruler-shipped-f95f2f70-c1-sampled

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	45.4 (45.4)	65.8 (67.0)	56.4 (55.9)	58.1 (55.3)	0.17-0.23 s

Ruler-shipped-f95f2f70-c1

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	47.5 (46.5)	69.5 (69.9)	58.2 (56.9)	60.2 (56.8)	0.16-0.23 s

Ruler-shipped-f95f2f70-c16

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
16	144.8 (8.9)	202.2 (12.3)	181.4 (11.4)	207.6 (12.5)	0.85-1.14 s

Ruler-shipped-f95f2f70-c8

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
8	111.1 (13.7)	154.1 (18.7)	139.5 (17.2)	157.3 (18.8)	0.45-0.60 s

Ablit/suite

check	passed
gsm8k	246/250
mmlu	292/342
humaneval	159/164
ifeval	118/150
tools	59/60
json	30/30
mgsm	221/240

Suite-ab

check	passed
gsm8k	244/250
mmlu	292/342
humaneval	156/164
ifeval	118/150
tools	59/60
json	30/30
mgsm	223/240

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-14 00:00	published evidence	published	68.7			1125/1236
2026-10-13 00:00	published evidence	published	1083.6			1874/2172
2026-10-12 00:00	published evidence	published	62.8	169.5	0.45-1.19 s	

GLM-5.3-Flash, MLX 4-bit on 1x Apple M5 Ultra (256 GB unified), macOS

Target glm53-m5: GLM-5.3-Flash MLX 4-bit, TensorFold native on one M5 Ultra Mac (one stream, greedy). Engine: TensorFold native, Metal; served as GLM-5.3-Flash-MLX-4bit. Latest: run evidence/2026-10-10-wiredreads (2026-10-10 00:00, plan published evidence, published).

Ruler-repeat

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	92.4 (92.4)	103.6 (103.6)	97.0 (97.0)	94.9 (94.9)	0.11-0.15 s

Ruler

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	90.7 (90.7)	103.7 (103.7)	99.1 (99.1)	41.8 (41.8)	0.12-0.15 s

Suite

check	passed
snapshot	12/12
drafts	12/12
concurrent	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-10 00:00	published evidence	published	103.7			36/36
2026-10-10 00:00	published evidence	published				1128/1213
2026-10-10 00:00	published evidence	published				1137/1213
2026-10-10 00:00	published evidence	published				1079/1213
2026-10-09 00:00	published evidence	published	104.6			
2026-10-09 00:00	published evidence	published				349/403
2026-10-09 00:00	published evidence	published				42/42

NVIDIA Nemotron 3.5 Bolt 74B-A9B (24 GB: IQ2_XXS experts, Q8_0, NVFP4 Mamba) + vision on 1x NVIDIA RT

Target nemotron-bolt-5090: Nemotron 3.5 Bolt 74B-A9B ModelOpt IQ2_XXS/Q8_0/NVFP4, TensorFold native CUDA on one RTX 5090 (262144, two streams, vision, host prompt cache). Engine: TensorFold native, CUDA; served as nemotron-bolt. Latest: run 20261015-175621-full (2026-10-15 17:56, plan full, done).

Serving: recipe_commit f546a71; engine_commit b2ec378d; served_model {"object": "list", "data": [{"id": "nemotron-bolt", "object": "model", "created": 17920869

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	149.7 (149.7)	149.6 (149.6)	150.1 (150.1)	149.7 (149.7)	0.03-0.03 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	1.011	0.0		6.75	217

Ruler-after

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	149.3 (149.3)	149.2 (149.2)	149.7 (149.7)	149.3 (149.3)	0.03-0.03 s

Engine rounds (ruler-after):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	1.007	0.0		6.74	1781

Suite

check	passed
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	8/8
stop	3/3
thinking	10/10
copy	2/2

check	passed
max-tokens	3/3
tools	58/60
json	30/30
ifeval	120/150
gsm8k	241/250
mgsm	190/240
mmlu	279/342
humaneval	153/164
repetition	8/8
long-context	55/55
vision	12/12
soak	60/60
snapshot	12/12

Checks

check	result
memory-idle	measured: gpu: 28716 MiB, 32607 MiB
structured	pass: 18/18 obey; a bad regex -> 400
sampling	pass: 9/9 seeded replies equal on repeat; 0 errors
turns	pass: resumed == again == cold: True; cached 5832 of 5865; ttft resumed 0.053s, cold 0.134s
capacity	pass: 6/6 answered, 0 refused, 0 lost (no reply); a short request after: 200

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-15 17:56	full	done	149.6			1272/1437

Qwen3.8-27B, MLX 4-bit + DFlash2 on 1x NVIDIA RTX 5090 (32 GB)

Target qwen38-27b-5090: Qwen3.8-27B MLX 4-bit + DFlash2, TensorFold native CUDA on one RTX 5090 (fp8 KV, 262k, one stream, vision). Engine: TensorFold native, CUDA; served as qwen38-27b. Latest: run evidence/2026-10-14-merge-1.0.11 (2026-10-14 00:00, plan published evidence, published).

Bench_openai-64tok

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	-	-	-	-	0.10-0.10 s

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-14 00:00	published evidence	published				
2026-10-13 00:00	published evidence	published				
2026-10-12 00:00	published evidence	published				1229/1437
2026-10-12 00:00	published evidence	published				1289/1437
2026-10-12 00:00	published evidence	published				
2026-10-12 00:00	published evidence	published				

Qwen3.8-Flash-Next, MLX 6-bit on 1x Apple M5 Ultra (256 GB unified), macOS

Target qwen38-flashnext-mlx6bit-m5: Qwen3.8-Flash-Next MLX 6-bit, TensorFold native on one M5 Ultra. Engine: TensorFold native, Metal; served as Qwen3.8-Flash-Next-MLX-6bit. Latest: run evidence/2026-10-11-suite-rest (2026-10-11 00:00, plan published evidence, published).

Suite

check	passed
ifeval	134/150
gsm8k	242/250
mgsm	225/240
mmlu	282/342
repetition	8/8
long-context	55/55
concurrent	12/12
snapshot	12/12

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-11 00:00	published evidence	published				970/1069
2026-10-11 00:00	published evidence	published	205.9			24/24
2026-10-11 00:00	published evidence	published				157/164
2026-10-11 00:00	published evidence	published	199.8			
2026-10-11 00:00	published evidence	published				131/132
2026-10-11 00:00	published evidence	published	206.2			

Qwen3.8-Flash-Next, 6-bit with vision on 1x Apple M5 Ultra (256 GB unified), macOS

Target qwen38-flashnext-vision-m5: Qwen3.8-Flash-Next 6-bit with image input, TensorFold native on one M5 Ultra. Engine: TensorFold native, Metal; served as Qwen3.8-Flash-Next-6bit. Latest: run evidence/2026-10-12-parallel8 (2026-10-12 00:00, plan published evidence, published).

Ruler

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	127.7 (127.7)	174.1 (174.1)	147.8 (147.8)	175.4 (175.4)	0.16-0.17 s
4	210.2 (52.9)	303.3 (76.3)	249.2 (61.1)	316.9 (77.8)	0.37-0.40 s
8	271.5 (34.1)	330.0 (40.8)	292.5 (36.5)	339.3 (41.7)	0.76-0.84 s

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-12 00:00	published evidence	published	174.1			
2026-10-12 00:00	published evidence	published	202.7			
2026-10-12 00:00	published evidence	published	176.6			
2026-10-11 00:00	published evidence	published	205.6			131/132
2026-10-11 00:00	published evidence	published	178.9			513/544

Qwen3.8-Flash-Next INT4-Mixed (AutoRound) on 2x NVIDIA DGX Spark (GB10, 128 GB unified each), TP=2 ov

Target qwen38-int4mixed: Qwen3.8-Flash-Next INT4-Mixed, TensorFold native, TP=2 on two DGX Sparks. Engine: TensorFold native, CUDA; served as Qwen3.8-Flash-Next-INT4-Mixed. Latest: run 20261015-171328-full (2026-10-15 17:13, plan full, done).

Serving: patch_sha 186193c6a530d89ee6383c11a9e41f98bac95bb119959aa5901c65b0f93b21ea; image tf-qwen38-int4mixed:native-v1.0.11-3 sha256:7c5838987336819a3dcc51d8f9abcf169f80fbfe73222e; server_args --context 1048576 --no-thinking --parallel 16 --vision-urls --vision

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	77.3 (77.3)	127.0 (127.0)	100.6 (100.6)	129.6 (129.6)	0.03-0.04 s
4	159.1 (39.8)	253.9 (65.0)	201.7 (49.8)	268.4 (61.8)	0.07-0.10 s
8	228.9 (28.8)	385.2 (46.2)	282.4 (35.8)	423.1 (51.3)	0.11-0.14 s
16	301.9 (18.8)	509.8 (30.6)	391.3 (24.0)	519.2 (31.6)	0.44-0.55 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.621	4.042	66%	33.33	12
4	3.173	2.949	74%	58.64	49
8	3.09	2.653	80%	76.26	100
16	3.222	2.925	77%	122.65	196

Ruler-after

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	77.5 (77.5)	125.9 (125.9)	99.2 (99.2)	129.7 (129.7)	0.04-0.04 s
16	302.3 (18.8)	509.5 (30.4)	389.8 (23.8)	520.4 (31.6)	0.44-0.50 s

Engine rounds (ruler-after):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	3.567	4.016	65%	33.01	13
16	3.21	2.914	76%	122.7	193

Suite

check	passed
drafts	12/12
concurrent	12/12
streaming	4/4
multiturn	8/8
stop	3/3
thinking	10/10
copy	2/2
max-tokens	3/3
tools	60/60
json	30/30
ifeval	130/150
gsm8k	241/250
mgsm	219/240
mmlu	283/342
humaneval	157/164
repetition	8/8
long-context	55/55
vision	12/12
soak	60/60
snapshot	12/12

Checks

check	result
memory-idle	measured: head 21.8 GiB available; worker 25.2 GiB available
structured	pass: 18/18 obey; a bad regex -> 400
sampling	pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors
turns	pass: resumed == again == cold: True; cached 5575 of 5642; ttft resumed 0.118s, cold 0.035s
urls	pass: public 200; loopback 400; http 400

check	result
capacity	pass: 4/6 answered, 2 refused, 0 lost (no reply); a short request after: 200

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-15 17:13	full	done	127.0	509.8	0.44-0.55 s	1321/1437
2026-10-15 00:00	published evidence	published	123.4	472.9	0.52-0.74 s	1321/1437
2026-10-15 00:00	published evidence	published				1321/1437
2026-10-11 00:00	published evidence	published	120.1	507.2	0.54-0.62 s	1321/1437
2026-10-11 00:00	published evidence	published	120.4	507.1	0.50-0.59 s	
2026-10-10 09:45	quick	done	105.3			54/54
2026-10-10 00:00	published evidence	published	122.0	510.3	0.40-0.49 s	114/114
2026-10-10 00:00	published evidence	published	125.9	510.9	0.44-0.49 s	1261/1377
2026-10-10 00:00	published evidence	published	125.7	506.7	0.42-0.49 s	54/54
2026-10-10 00:00	published evidence	published	106.1	506.1	0.42-0.51 s	54/54
2026-10-10 00:00	published evidence	published	110.0	511.8	0.41-0.55 s	1261/1377
2026-10-10 00:00	published evidence	published	105.8	505.5	0.41-0.46 s	1261/1377
2026-10-09 00:00	published evidence	published	105.2	412.2	0.78-0.98 s	1260/1377
2026-10-09 00:00	published evidence	published				1256/1377
2026-10-08 00:00	published evidence	published	107.1	519.4	0.44-0.56 s	
2026-10-08 00:00	published evidence	published	107.7	474.0	0.52-0.66 s	145/145
2026-10-08 00:00	published evidence	published	107.6	442.6	0.44-0.58 s	36/36
2026-10-08 00:00	published evidence	published	107.6	441.7	0.45-0.58 s	
2026-10-07 00:00	published evidence	published				12/12

Step-5-Preview, MLX 2-bit experts (8-bit rest) + MTP on 1x Apple M5 Ultra (256 GB unified), macOS

Target step5-m5: Step-5-Preview 2-bit (experts 2/128, rest 8/64, MTP, images), TensorFold native Metal on one M5 Ultra. Engine: TensorFold native, Metal; served as step5-preview. Latest: run 20261015-194421-one-stream (2026-10-15 19:44, plan one-stream, done).

Serving: engine_commit 6e791da6; binary_sha256 be201a04b22bc474aa9f61091078dd7d07354b61683cd1ac0eed12374e3ddaab; served_model {"object": "list", "data": [{"id": "step5-preview", "object": "model", "created": 17921115; chip Apple M5 Ultra; 274877906944

Ruler-fresh

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	39.3 (39.3)	50.9 (50.9)	40.7 (40.7)	48.3 (48.3)	0.24-0.28 s

Engine rounds (ruler-fresh):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	2.381	2.072	67%	53.15	12

Ruler-after

streams	prose total (a stream)	code total (a stream)	structured total (a stream)	list total (a stream)	first token
1	39.1 (39.1)	50.6 (50.6)	40.6 (40.6)	48.2 (48.2)	0.24-0.28 s

Engine rounds (ruler-after):

streams	tokens a round	drafts a round	accepted	ms a round	requests
1	2.339	1.939	69%	52.39	13

Suite

check	passed
drafts	12/12
streaming	4/4
multiturn	8/8
stop	3/3
thinking	10/10
copy	2/2

check	passed
max-tokens	3/3
tools	58/60
json	30/30
gsm8k	49/50
ifeval	41/50

Checks

check	result
memory-idle	measured: mac: System-wide memory free percentage: 27%; total = 4096.00M us
sampling	pass: 9/9 seeded replies equal on repeat and undrafted; 0 errors
memory-busy	measured: mac: System-wide memory free percentage: 27%; total = 4096.00M us

History

run	plan	status	1 stream code	16 streams code	16 streams first token	suite
2026-10-15 19:44	one-stream	done	50.9			220/232
2026-10-15 19:28	quick	done	50.5			114/114
2026-10-15 13:43	one-stream	done	50.6			220/232

Served earlier, numbers in their READMEs only

These recipes predate the ruler's JSON, so their measurements are not in the ledger; their READMEs hold them.

model	hardware	engine	recipe
Qwen3.8-Flash-Next, turboderp EXL3 4.05 bpw	2x NVIDIA DGX Spark (GB10), TP=2	TensorFold 0.6.6 (Python) with an EXL3 patch	github.com/BobClawblaw/qwen38-flashnext-exl3-2x-dgx-sparks
Qwen3.8-Flash-Next (TensorFold 0.6.6 recipe)	2x NVIDIA DGX Spark (GB10), TP=2	TensorFold 0.6.6 (Python)	github.com/BobClawblaw/qwen38-flashnext-tensorfold-2x-dgx-sparks

Step-5-Preview on one M5 Ultra, final build: measured outside the harness

TensorFold native Metal, branch step5-native at engine 6e791da6: 2,048-row prompt chunks, the prompt cache and the idle keepalive's kernel tick. Greedy unless said; served with --parallel 2 --vision at the default context (227,025).

what	measured
prompt, 4k / 8k / 16k / 32k tokens	868 / 863 / 840 / 800 tok/s (a 4k prompt right after a pause was 580 before the keepalive fix)
212k-token chat question, one record's code of 10,500	2 of 3 right (records 3,000 and 9,800; missed 777), 376 s each, at least 19% of memory free
129k-token chat question	2 of 2 right (records 777 and 5,000), about 200 s each
MTP drafts against "draft": false, 200-400-token replies timed with their prompt	identical; code 25.8 to 41.2 tok/s, prose 27.0 to 36.6, math 26.9 to 48.8
prompt cache, an 11.6k-token chat sent again, then two more turns	15.3 s fresh, 1.8 s again, turns resumed at 11,601; replies identical to --prompt-cache-gib 0
prompt cache, images (1200 x 800 PNG, 688 tokens)	resent in 2.15 s instead of 3.8 s; replies identical to the cache off
a kept state that cannot sit beside its stream (2.4 GiB cache budget)	dropped before the pass; the prompt read again with the same reply
a 1-token reply after 2 / 6 / 30 / 120 s idle	0.17 / 0.18 / 0.18 / 0.19 s (about 3 s before the kernel tick and residency set)
two 18k/31k-token chats at once	both right

Step-5-Preview on one M5 Ultra, first build (512-row chunks): measured outside the harness

TensorFold native Metal, branch step5-native (engine 5ad7660c, images aa221b11), MLX 2-bit experts with the rest 8-bit, greedy, one stream. The harness's step5-m5 runs hold the ruler and the suite; these are the rest.

what	measured
prompt, 4k / 8k / 16k / 32k tokens	690 / 668 / 647 / 622 tok/s (the sparse indexer caps each row's keys past 4k)
131,016-token chat question about its first page	answered exactly, 264 s end to end, at least 23% of memory free
decode without drafts, 20 / 600 / 4,200 / 32,768 tokens of context	29.6 / 21.9 / 14.7 / 14.5 tok/s
MTP drafts against "draft": false (replies token-identical)	code 25.7 to 44.6 tok/s (70% accepted), prose 26.4 to 34.8 (56%), math 26.2 to 46.0 (75%)
an image, 1200 x 800 PNG (six 504-px crops and the overview, 692 prompt tokens)	2.1 s to a 4-token answer
an image, 600 x 400 PNG (overview only, 193 prompt tokens)	0.7 s